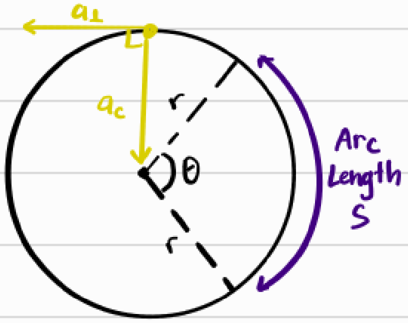
Circular motion



Linear velocity, v : Resultant of all straight line velocity components
Radial velocity, v_r : How fast the object moves along the position vector Perp. velocity, $v_\perp$: How fast the object moves along the curve

$$v = \sqrt{v_r^2 + v_\perp^2}$$

① Formula relating S & θ is

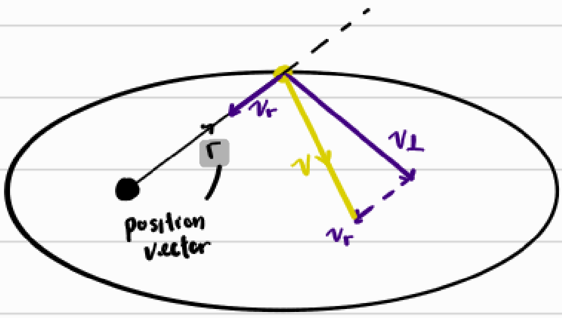
$$\boxed{S = r\theta}$$

Taking $\frac{d}{dt}$ on both sides

$$\Longrightarrow \frac{ds}{dt} = r\frac{d\theta}{dt}$$

$$\boxed{v_\perp = r\omega} \qquad \boxed{v\sin\theta = r\omega}$$

Perp. Velocity [Perp. to r] — Angular velocity — Angle btwn v & r



Taking $\frac{d}{dt}$ on both sides again

Constant

$$\frac{dv_\perp}{dt} = r\frac{d\omega}{dt}$$

$$\boxed{a_\perp = r\alpha}$$

Perp. Acceleration — Angular acceleration

[Radial acceleration]
Centripetal acceleration: $\boxed{a_c = \frac{v_\perp^2}{r} = \frac{(r\omega)^2}{r} = r\omega^2}$

Linear velocity : How fast an object is moving along the circumference of the circle

Angular velocity : How fast an object can cover 360°/2π rad.

* For a rigid body, the angular quantities will be constant [ω, α] since every particle finishes 360° at the same time. The linear quantities [v, a_t, a_c] will be diff for diff points for diff r.

Rotational KE & Rotational Inertia

Derivation of Rotational Inertia, I

For linear motion, $F = ma_t = m(r\alpha)$ —— ①

For rotational motion, $\tau = I\alpha$ (Angular Acceleration) —— ②

Torque, $\tau = Fr$ —— ③

Newton 2nd Law [Linear / Rotation]

Recap:

$F = ma$ (m: measure of inertia)

$\Longrightarrow m = \frac{F}{a}$

This formula shows that a larger mass, m requires a larger force to achieve the same acceleration

③ → ②

$Fr = I\alpha$

$I = \frac{Fr}{\alpha}$ —— ④

① → ④

$I = \frac{(mr\alpha)(r)}{\alpha}$

$= mr^2$ [For a point mass]

For a system of particles,

$\boxed{I = \sum m_i r_i^2}$

r_i: Perp. dist from point mass to rotational axis [old axis]

* Rotational Inertia is unique for diff rotational axis / pivot

Rotational KE of one particle

$KE_i = \frac{1}{2} m_i v_i^2 = \frac{1}{2} m_i (r_i w)^2 = \frac{1}{2} m_i r_i^2 w_i^2$

Rotational KE of all particles (Entire Object)

$\sum KE = \sum \frac{1}{2} m_i r_i^2 w^2 = \frac{1}{2} w^2 \sum m_i r_i^2$

$\boxed{KE_{rotational} = \frac{1}{2} w^2 I}$

* $KE_{Rot} = \sum KE_{linear}$ of all particles in an object, compactly expressed using rotational inertia to avoid calculating infinite individual terms

Parallel axis theorem [Reference point : CM]

Let the new axis, r_i' be d away from Centre of mass [CM]

$r_i' = r_i + d$

So the total rotational inertia is,

$I_{new} = \sum m_i [r_i + d]^2$

$= \sum m_i [r_i^2 + 2r_i d + d^2]$

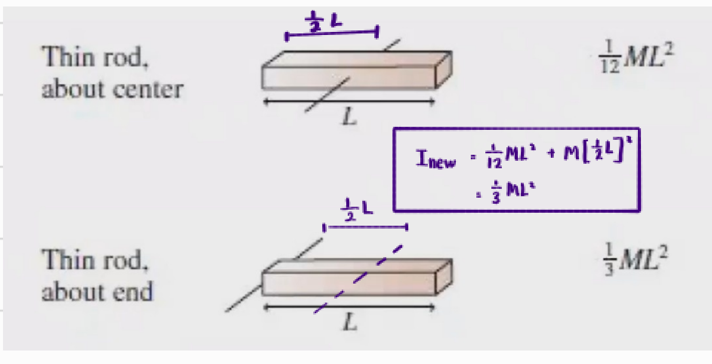
$= \sum m_i r_i^2 + 2d \sum m_i r_i + d^2 \sum m_i$

$= I_{cm} + 0 + d^2 M$

$\boxed{I_{new} = I_{cm} + Md^2}$

d: Dist. btwn old axis [CM] & new axis

Since our ref. point is at CM, $\sum m_i r_i$ at CM $= 0$ bcuz all moments cancel out. That's why we can balance things when we put our finger at the CM



Work done by constant torque

$$\boxed{W = \tau \Delta\theta}$$

Angular displacement

Power done by constant torque

$$\frac{dW}{dt} = \frac{d}{dt}[\tau \Delta\theta]$$

constant

$$= \tau \left[\frac{d\theta}{dt}\right]$$

$$\boxed{P = \tau \omega}$$

Conservation of total angular momentum, L

Linear, $p = mv$

$\boxed{\text{Angular, } L = I\omega}$

$= \quad L = (mr^2)\left(\frac{v}{r}\right)$

$\quad\quad = r(mv)$

$\boxed{L = rp}$

Linear Motion	Rotational Motion	Meaning
Displacement x	Angular Displacement θ	Position in space vs. angle rotated
Velocity v	Angular Velocity ω	Rate of change of position vs. rotation speed
Acceleration a	Angular Acceleration α	Rate of change of velocity vs. how fast rotation changes
Mass m	Moment of Inertia I	Measure of inertia (resistance to motion)
Force F	Torque τ	Push/Pull vs. Rotational force
Linear Momentum $p = mv$	Angular Momentum $L = I\omega$	"Quantity of motion" for linear vs. rotational motion
Newton's 2nd Law $F = ma$	Rotational 2nd Law $\tau = I\alpha$	Relates force & acceleration to torque & angular acceleration
Kinetic Energy $KE = \frac{1}{2}mv^2$	Rotational KE $KE_{\text{rot}} = \frac{1}{2}I\omega^2$	Energy of motion vs. energy of spinning
Work Done $W = Fs$	Rotational Work $W = \tau\theta$	Force over distance vs. torque over angle
Power $P = Fv$	Rotational Power $P = \tau\omega$	Rate of doing work in linear vs. rotational systems
Impulse $J = F\Delta t$	Angular Impulse $J_{\text{rot}} = \tau\Delta t$	Change in momentum for linear vs. rotational motion

Linear & Rotational counterparts

Displacement, s $\leftrightarrow$ Angular displacement, $\Delta\theta$

Velocity, v $\leftrightarrow$ Angular velocity, ω

Acceleration, a $\leftrightarrow$ Angular acceleration, α

mass, m $\leftrightarrow$ Rotational Inertia, I

Force, F $\leftrightarrow$ Torque, τ

Common Moments of Inertia